

Muskan Chawla

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SUMMARY

Backend-focused software engineer building scalable, production-oriented systems with strong foundations in Node.js, NestJS, TypeScript, PostgreSQL, and system design. Experienced in developing real-time architectures, secure APIs, and deployment-focused backend workflows with an emphasis on performance, scalability, and problem-solving. Passionate about engineering reliable software systems, strengthening core computer science fundamentals, and solving complex technical challenges.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript, SQL

Backend Development: Node.js, NestJS, REST APIs, WebSockets, JWT Authentication, API Security

Databases & Caching: PostgreSQL, Redis, Database Design, Query Optimization

DevOps & Tools: Docker, Git, GitHub, Postman, CI/CD Basics

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, System Design, Scalability, Asynchronous Architecture

PROJECTS

Real-Time Multiplayer Drawing & Guessing Platform

GitHub – Live Demo

NestJS, Node.js, WebSockets, PostgreSQL, JWT, Redis

- Engineered a real-time multiplayer platform enabling synchronized drawing, live guessing, scoring, and room-based gameplay through WebSocket-driven event architecture
- Designed server-authoritative game state management with room lifecycle control, player rotation, reconnect handling, and low-latency event synchronization
- Implemented secure authentication, payload validation, anti-spam throttling, and scalable multiplayer session workflows

Containerized Deployment Automation Platform (Vercel Lite)

GitHub – Live Demo

NestJS, Docker, BullMQ/Redis, PostgreSQL, Cloud Storage

- Architected a deployment automation platform supporting repository/ZIP ingestion, isolated builds, deployment lifecycle orchestration, and hosted static asset delivery
- Developed asynchronous queue-based deployment pipelines with worker processing, retry systems, timeout controls, and deployment state management
- Implemented Dockerized build isolation, artifact versioning, secure execution workflows, and rollback-ready deployment infrastructure

EXPERIENCE

AI Research Intern

June 2022 – Aug 2022

DeepMind

London, UK

- Conducted research on reinforcement learning algorithms for robotics
- Implemented and evaluated deep RL models using PyTorch and RLlib
- Presented findings at weekly research meetings

Machine Learning Engineer

Jan 2021 – May 2022

Acme AI Solutions

San Francisco, CA

- Developed and deployed machine learning models for various industries
- Optimized model performance and ensured data quality
- Collaborated with cross-functional teams to deliver AI solutions

EDUCATION

Chitkara University

B.E. in Computer Science Engineering

Punjab, India

May 2022 – Present